

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbially	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 13 (29 Nov-3 Dec 2026) - Period 2	Unit 4 - 4.3 Game Enhancements (2/2)
Pacing (2-1-1-2): ☒ On plan ☐ Ahead ☐ Behind Weeks ahead / behind: 0 Week / lesson: Week 13 / Lesson 2 of 2		Starting point (2-2-1-3): What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it? Popups working for 70%; z-index errors common. Today: quick fix-it and add audio elements and sound effects.		

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — trace to the curriculum / standard and the table of specifications (ToS)

Students implement audio elements (background music with play/pause/volume) and sound effects on events (catch, miss, end), considering user control and accessibility. Sprint 4.3 obj. 3-4; Challenges 1, 3; CSTA 3A-AP-21.

Success criteria — what will successful learning look / sound like?

I add an audio element and control it with JS (play, pause, volume); sound effects play on catch and game over; players can mute.

Key knowledge / skill / understanding:

HTML audio element; Audio() object; browsers block autoplay until user interaction.

Likely misconception / access or challenge need:

Music autoplays on load; louder sound = better game.

2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 student task + evidence of learning	ACTIVITY 2 student task + evidence of learning	ACTIVITY 3 student task + evidence of learning
Strategy: Inquiry / investigation ☐ Time: 8 min	Strategy: Problem solving ☐ Time: 20 min	Strategy: Role play ☐ Time: 9 min
What students will do: Investigate the audio element: test controls, autoplay, loop, volume in the browser; find out why autoplay is blocked (MDN).	What students will do: Challenge 1: build a simple audio player (play/pause/volume); then add catch, miss and game-over sounds plus a mute button to your game.	What students will do: Pairs play 'player and developer': player gives feedback on sound experience (too loud, annoying) and developer adjusts.
Evidence tool: Observation checklist ☐ Assessment: ☒ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group	Evidence tool: Practical / performance task ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☒ Individual ☐ Pair ☐ Group	Evidence tool: Peer-assessment checklist ☐ Assessment: ☐ Diagnostic ☒ Formative ☐ End Grouping: ☐ Individual ☒ Pair ☐ Group
Resources: MDN Web Docs (trusted source); test page.	Resources: VS Code; audio files from project; QR 'Audio elements'.	Resources: Feedback checklist.
Support (low performers / SEN): Guided question sheet.	Support (low performers / SEN): Player HTML provided; add 2 sounds only.	Support (low performers / SEN): Feedback phrases card.
Stretch (fast learners / gifted): Explain browser autoplay policy and why it exists.	Stretch (fast learners / gifted): Challenge 3: image carousel with Next/Previous.	Stretch (fast learners / gifted): Developer writes a change log entry.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

- ☒ Hook / prior learning ☒ Explain or model ☒ Success criteria shared
- ☒ Key vocabulary / examples ☒ Real-life connection ☒ Clear instructions / resources
- ☒ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Fix-it: 3-minute z-index demo. Hook: game with and without sound. Model new Audio(), play(), pause(), volume, and attaching a sound to the catch event. Remind headphones and respectful volume.

What is new or innovative in this lesson? (ETEC looks for this)

Students use MDN as a trusted source (digital research) and apply user-centred feedback via player/developer role play.

Progress recorded / shared:

- ☒ Gradebook ☐ Parents ☒ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

- ☒ Re-teach ☒ Regroup ☒ Give extra support
- ☒ Stretch ☒ Have students redo / improve

What I will watch for / likely responses:

Autoplay errors in console; wrong audio path; sounds overlapping (reset currentTime). Response: 'no sound' - path or blocked autoplay.

Close / exit check (summative + time for learners' questions) Time: 7 min

Exit check: why must audio start after a user click? Write the line that sets volume to 50%. Learners' questions.

Homework / platform (e.g. Madrasati)

Madrasati: upload game with sounds (zip). Prepare one animation idea for the worms or gloves.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

MDN investigation built research habits; role play gave sensible sound choices. Students were considerate about music choices for peers from different backgrounds.

What will I change next time?

Have headphones for every PC; noise made testing hard. Pre-check audio file paths.

Students needing follow-up:

Students missing audio files in their folders (approx. 2 per class). Gifted students finished carousel - use as showcase helpers.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☒ 1 ☐ 2 ☐ 3 ☒ C ☐ H	☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☒ 1 ☒ 2 ☐ 3 ☐ C ☒ H	☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☐ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H
Create something new · 2-1-1-7	Discussion expressing ideas · 2-1-1-8	Pair / group cooperative work · 2-1-1-8	Learner choice independent work · 3-2-1-6	Self-assessment 2-2-1-1	Peer assessment 2-2-1-1	SEN / low specific support · 2-1-1-3	Gifted specific stretch · 2-1-1-3
☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☐ 1 ☒ 2 ☐ 3 ☐ C ☐ H	☐ 1 ☐ 2 ☐ 3 ☒ C ☐ H	☐ 1 ☐ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H	☒ 1 ☒ 2 ☒ 3 ☐ C ☐ H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback